

```
import zipfile
import os

zip_path = "/content/archive.zip"
extract_path = "/content/olist_data"

with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall(extract_path)

    print("Dataset extracted successfully!")
    print("\nFiles available:")

    for file in os.listdir(extract_path):
        print(file)
```

Dataset extracted successfully!

Files available:

- olist_order_items_dataset.csv
- olist_products_dataset.csv
- product_category_name_translation.csv
- olist_sellers_dataset.csv
- olist_order_reviews_dataset.csv
- olist_customers_dataset.csv
- olist_orders_dataset.csv
- olist_geolocation_dataset.csv
- olist_order_payments_dataset.csv

```
status_percent = orders["order_status"].value_counts(normalize=True) * 100

print("Order Status Percentage:")
print(status_percent.round(2))
```

Order Status Percentage:

order_status	
delivered	97.02
shipped	1.11
canceled	0.63
unavailable	0.61
invoiced	0.32
processing	0.30
created	0.01
approved	0.00

Name: proportion, dtype: float64

```
print("Average Customer Review Score:")
print(round(reviews["review_score"].mean(), 2))
```

Average Customer Review Score:
4.09

```
print("Payment Value Analysis")

print("Total Payment Value:", round(payments["payment_value"].sum(), 2))
print("Average Payment Value:", round(payments["payment_value"].mean(), 2))
```

Payment Value Analysis
Total Payment Value: 16008872.12
Average Payment Value: 154.1

```
payment_revenue = payments.groupby("payment_type")["payment_value"].sum()
payment_revenue = payment_revenue.sort_values(ascending=False)
```

```
print("Revenue by Payment Method:")
print(payment_revenue.round(2))
```

```
Revenue by Payment Method:
payment_type
credit_card      12542084.19
boleto           2869361.27
voucher          379436.87
debit_card       217989.79
not_defined      0.00
Name: payment_value, dtype: float64
```

```
print("Business Summary")
```

```
print("Total Orders:", orders["order_id"].nunique())
print("Total Items Sold:", len(items))
print("Total Revenue:", round(items["price"].sum(), 2))
print("Average Order Value:", round(items["price"].sum() / orders["order_id"].nunique(), 2))
```

```
Business Summary
Total Orders: 99441
Total Items Sold: 112650
Total Revenue: 13591643.7
Average Order Value: 136.68
```

```
import pandas as pd, os
path="/content/olist_data"
[(print(f, "->", pd.read_csv(os.path.join(path,f)).shape)) for f in os.listdir(path) if
```

```
olist_order_items_dataset.csv -> (112650, 7)
olist_products_dataset.csv -> (32951, 9)
product_category_name_translation.csv -> (71, 2)
olist_sellers_dataset.csv -> (3095, 4)
olist_order_reviews_dataset.csv -> (99224, 7)
olist_customers_dataset.csv -> (99441, 5)
olist_orders_dataset.csv -> (99441, 8)
olist_geolocation_dataset.csv -> (1000163, 5)
olist_order_payments_dataset.csv -> (103886, 5)
[None, None, None, None, None, None, None, None, None]
```

```
("PROJECT SUMMARY")
("-----")
("Total Orders:", orders["order_id"].nunique())
("Total Items Sold:", len(items))
("Total Revenue:", round(items["price"].sum(), 2))
("Average Order Value:", round(items["price"].sum() / orders["order_id"].nunique(), 2))
```

```
PROJECT SUMMARY
-----
Total Orders: 99441
Total Items Sold: 112650
Total Revenue: 13591643.7
Average Order Value: 136.68
```

```
print(orders.shape)
print(orders.columns.tolist())
print(orders.head(2))
```

```
(99441, 8)
['order_id', 'customer_id', 'order_status', 'order_purchase_timestamp', 'order_approved_at',
 'order_id', 'customer_id']
0  e481f51cbdc54678b7cc49136f2d6af7  9ef432eb6251297304e76186b10a928d
1  53cdb2fc8bc7dce0b6741e2150273451  b0830fb4747a6c6d20dea0b8c802d7ef

order_status order_purchase_timestamp order_approved_at \
0  delivered      2017-10-02 10:56:33  2017-10-02 11:07:15
1  delivered      2018-07-24 20:41:37  2018-07-26 03:24:27
```

	order_delivered_carrier_date	order_delivered_customer_date	\
0	2017-10-04 19:55:00	2017-10-10 21:25:13	
1	2018-07-26 14:31:00	2018-08-07 15:27:45	

	order_estimated_delivery_date
0	2017-10-18 00:00:00
1	2018-08-13 00:00:00

```
# Calculate total revenue
total_revenue = items["price"].sum()

# Total orders
total_orders = orders["order_id"].nunique()

# Total products sold
total_items = items["order_id"].count()

# Average order value
average_order_value = total_revenue / total_orders

print("Total Revenue:", round(total_revenue, 2))
print("Total Orders:", total_orders)
print("Total Items Sold:", total_items)
print("Average Order Value:", round(average_order_value, 2))
```

```
Total Revenue: 13591643.7
Total Orders: 99441
Total Items Sold: 112650
Average Order Value: 136.68
```

```
import pandas as pd
import os

path = "/content/olist_data"

orders = pd.read_csv(path + "/olist_orders_dataset.csv")
items = pd.read_csv(path + "/olist_order_items_dataset.csv")
products = pd.read_csv(path + "/olist_products_dataset.csv")
customers = pd.read_csv(path + "/olist_customers_dataset.csv")
sellers = pd.read_csv(path + "/olist_sellers_dataset.csv")
payments = pd.read_csv(path + "/olist_order_payments_dataset.csv")
reviews = pd.read_csv(path + "/olist_order_reviews_dataset.csv")
category = pd.read_csv(path + "/product_category_name_translation.csv")

print("All datasets loaded successfully!")
print("Orders:", orders.shape)
print("Items:", items.shape)
print("Products:", products.shape)
print("Customers:", customers.shape)
print("Sellers:", sellers.shape)
print("Payments:", payments.shape)
print("Reviews:", reviews.shape)
```

```
All datasets loaded successfully!
Orders: (99441, 8)
Items: (112650, 7)
Products: (32951, 9)
Customers: (99441, 5)
Sellers: (3095, 4)
Payments: (103886, 5)
Reviews: (99224, 7)
```

```
print(orders["order_purchase_timestamp"].head())
```

```
0    2017-10-02 10:56:33
1    2018-07-24 20:41:37
2    2018-08-08 08:38:49
3    2017-11-18 19:28:06
4    2018-02-13 21:18:39
Name: order_purchase_timestamp, dtype: object
```

```
print("Top 10 Most Expensive Products")

top_products = items.groupby("product_id")["price"].sum()
top_products = top_products.sort_values(ascending=False)

print(top_products.head(10))
```

```
Top 10 Most Expensive Products
product_id
bb50f2e236e5eea0100680137654686c    63885.00
6cdd53843498f92890544667809f1595    54730.20
d6160fb7873f184099d9bc95e30376af    48899.34
d1c427060a0f73f6b889a5c7c61f2ac4    47214.51
99a4788cb24856965c36a24e339b6058    43025.56
3dd2a17168ec895c781a9191c1e95ad7    41082.60
25c38557cf793876c5abdd5931f922db    38907.32
5f504b3a1c75b73d6151be81eb05bdc9    37733.90
53b36df67ebb7c41585e8d54d6772e08    37683.42
aca2eb7d00ea1a7b8ebd4e68314663af    37608.90
Name: price, dtype: float64
```

```
print(orders["order_status"].value_counts())
```

```
order_status
delivered    96478
shipped      1107
canceled      625
unavailable   609
invoiced      314
processing    301
created        5
approved       2
Name: count, dtype: int64
```

```
print(customers["customer_state"].value_counts().head(10))
```

```
customer_state
SP    41746
RJ    12852
MG    11635
RS     5466
PR     5045
SC     3637
BA     3380
DF     2140
ES     2033
GO     2020
Name: count, dtype: int64
```

```
print(payments["payment_type"].value_counts())
```

```
payment_type
credit_card    76795
boleto        19784
voucher        5775
debit_card     1529
not_defined      3
Name: count, dtype: int64
```

```
print("Customer Review Scores")
print(reviews["review_score"].value_counts().sort_index())
```

```
Customer Review Scores
review_score
1      11424
2       3151
3       8179
4      19142
5      57328
Name: count, dtype: int64
```

```
print("Top Seller States")
print(sellers["seller_state"].value_counts().head(10))
```

```
Top Seller States
seller_state
SP       1849
PR        349
MG        244
SC        190
RJ        171
RS        129
GO         40
DF         30
ES         23
BA         19
Name: count, dtype: int64
```

```
print("Payment Type Analysis")
print(payments["payment_type"].value_counts())
```

```
Payment Type Analysis
payment_type
credit_card      76795
boleto          19784
voucher          5775
debit_card       1529
not_defined         3
Name: count, dtype: int64
```

```
print(products["product_category_name"].value_counts().head(10))
```

```
product_category_name
cama_mesa_banho      3029
esporte_lazer        2867
moveis_decoracao     2657
beleza_saude         2444
utilidades_domesticas 2335
automotivo           1900
informatica_acessorios 1639
brinquedos          1411
relogios_presentes   1329
telefonica           1134
Name: count, dtype: int64
```

```
print("ORDERS")
print(orders.head())

print("\nORDER ITEMS")
print(items.head())

print("\nPAYMENTS")
print(payments.head())
```

1	delivered	2018-07-24 20:41:37	2018-07-26 03:24:27
2	delivered	2018-08-08 08:38:49	2018-08-08 08:55:23
3	delivered	2017-11-18 19:28:06	2017-11-18 19:45:59
4	delivered	2018-02-13 21:18:39	2018-02-13 22:20:29

	order_delivered_carrier_date	order_delivered_customer_date	\
0	2017-10-04 19:55:00	2017-10-10 21:25:13	
1	2018-07-26 14:31:00	2018-08-07 15:27:45	
2	2018-08-08 13:50:00	2018-08-17 18:06:29	
3	2017-11-22 13:39:59	2017-12-02 00:28:42	
4	2018-02-14 19:46:34	2018-02-16 18:17:02	

	order_estimated_delivery_date
0	2017-10-18 00:00:00
1	2018-08-13 00:00:00
2	2018-09-04 00:00:00
3	2017-12-15 00:00:00
4	2018-02-26 00:00:00

ORDER ITEMS

	order_id	order_item_id	\
0	00010242fe8c5a6d1ba2dd792cb16214	1	
1	00018f77f2f0320c557190d7a144bdd3	1	
2	000229ec398224ef6ca0657da4fc703e	1	
3	00024acbcdcf0a6daa1e931b038114c75	1	
4	00042b26cf59d7ce69dfabb4e55b4fd9	1	

	product_id	seller_id	\
0	4244733e06e7ecb4970a6e2683c13e61	48436dade18ac8b2bce089ec2a041202	
1	e5f2d52b802189ee658865ca93d83a8f	dd7ddc04e1b6c2c614352b383efe2d36	
2	c777355d18b72b67abbeef9df44fd0fd	5b51032eddd242adc84c38acab88f23d	
3	7634da152a4610f1595efa32f14722fc	9d7a1d34a5052409006425275ba1c2b4	
4	ac6c3623068f30de03045865e4e10089	df560393f3a51e74553ab94004ba5c87	

	shipping_limit_date	price	freight_value
0	2017-09-19 09:45:35	58.90	13.29
1	2017-05-03 11:05:13	239.90	19.93
2	2018-01-18 14:48:30	199.00	17.87
3	2018-08-15 10:10:18	12.99	12.79
4	2017-02-13 13:57:51	199.90	18.14

PAYMENTS

	order_id	payment_sequential	payment_type	\
0	b81ef226f3fe1789b1e8b2acac839d17	1	credit_card	
1	a9810da82917af2d9aefd1278f1dcfa0	1	credit_card	
2	25e8ea4e93396b6fa0d3dd708e76c1bd	1	credit_card	
3	ba78997921bbcdc1373bb41e913ab953	1	credit_card	
4	42fdf880ba16b47b59251dd489d4441a	1	credit_card	

	payment_installments	payment_value
0	8	99.33
1	1	24.39
2	1	65.71
3	8	107.78
4	2	128.45

```
print("Order Status Analysis")
print(orders["order_status"].value_counts())
```

```
Order Status Analysis
order_status
delivered      96478
shipped         1107
canceled         625
unavailable     609
invoiced        314
processing       301
created           5
approved         2
Name: count, dtype: int64
```

```
print("Top Customer States")
print(customers["customer_state"].value_counts().head(10))
```

```
Top Customer States
customer_state
SP      41746
RJ      12852
MG      11635
RS       5466
PR       5045
SC       3637
BA       3380
DF       2140
ES       2033
GO       2020
Name: count, dtype: int64
```

```
print("Review Score Distribution")
print(reviews["review_score"].value_counts().sort_index())
```

```
Review Score Distribution
review_score
1      11424
2       3151
3       8179
4      19142
5      57328
Name: count, dtype: int64
```

```
print("Order Status Distribution")
print(orders["order_status"].value_counts())
```

```
Order Status Distribution
order_status
delivered      96478
shipped         1107
canceled         625
unavailable      609
invoiced         314
processing       301
created           5
approved         2
Name: count, dtype: int64
```

```
print("Top Product Categories")
print(products["product_category_name"].value_counts().head(10))
```

```
Top Product Categories
product_category_name
cama_mesa_banho      3029
esporte_lazer        2867
moveis_decoracao     2657
beleza_saude         2444
utilidades_domesticas 2335
automotivo            1900
informatica_acessorios 1639
brinquedos           1411
relogios_presentes    1329
telefonica            1134
Name: count, dtype: int64
```

```
print("Payment Method Distribution")
print(payments["payment_type"].value_counts())
```

```
Payment Method Distribution
payment_type
credit_card      76795
boleto           19784
```

```
voucher          5775
debit_card       1529
not_defined       3
Name: count, dtype: int64
```

```
print("Data Quality Check")

print("Orders missing values:")
print(orders.isnull().sum())

print("\nProducts missing values:")
print(products.isnull().sum())

print("\nReviews missing values:")
print(reviews.isnull().sum())
```

```
Data Quality Check
Orders missing values:
order_id          0
customer_id       0
order_status      0
order_purchase_timestamp  0
order_approved_at 160
order_delivered_carrier_date 1783
order_delivered_customer_date 2965
order_estimated_delivery_date  0
dtype: int64

Products missing values:
product_id        0
product_category_name  610
product_name_lenght  610
product_description_lenght  610
product_photos_qty  610
product_weight_g    2
product_length_cm   2
product_height_cm   2
product_width_cm    2
dtype: int64

Reviews missing values:
review_id         0
order_id          0
review_score      0
review_comment_title  87656
review_comment_message  58247
review_creation_date  0
review_answer_timestamp  0
dtype: int64
```

```
print("Duplicate Records")

print("Orders duplicates:", orders.duplicated().sum())
print("Items duplicates:", items.duplicated().sum())
print("Products duplicates:", products.duplicated().sum())
print("Customers duplicates:", customers.duplicated().sum())
```

```
Duplicate Records
Orders duplicates: 0
Items duplicates: 0
Products duplicates: 0
Customers duplicates: 0
```

```
top_revenue = items.groupby("product_id")["price"].sum().sort_values(ascending=False).h

print("Top 10 Products by Revenue:")
print(top_revenue)
```

```

Top 10 Products by Revenue:
product_id
bb50f2e236e5eea0100680137654686c    63885.00
6cdd53843498f92890544667809f1595    54730.20
d6160fb7873f184099d9bc95e30376af    48899.34
d1c427060a0f73f6b889a5c7c61f2ac4    47214.51
99a4788cb24856965c36a24e339b6058    43025.56
3dd2a17168ec895c781a9191c1e95ad7    41082.60
25c38557cf793876c5abdd5931f922db    38907.32
5f504b3a1c75b73d6151be81eb05bdc9    37733.90
53b36df67ebb7c41585e8d54d6772e08    37683.42
aca2eb7d00ea1a7b8ebd4e68314663af    37608.90
Name: price, dtype: float64

```

```

print("Top 10 Products by Number of Sales")

top_sales = items["product_id"].value_counts().head(10)

print(top_sales)

```

```

Top 10 Products by Number of Sales
product_id
aca2eb7d00ea1a7b8ebd4e68314663af    527
99a4788cb24856965c36a24e339b6058    488
422879e10f46682990de24d770e7f83d    484
389d119b48cf3043d311335e499d9c6b    392
368c6c730842d78016ad823897a372db    388
53759a2ecddad2bb87a079a1f1519f73    373
d1c427060a0f73f6b889a5c7c61f2ac4    343
53b36df67ebb7c41585e8d54d6772e08    323
154e7e31ebfa092203795c972e5804a6    281
3dd2a17168ec895c781a9191c1e95ad7    274
Name: count, dtype: int64

```

```

print("Missing Values - Orders")
print(orders.isnull().sum())

print("\nMissing Values - Items")
print(items.isnull().sum())

print("\nMissing Values - Payments")
print(payments.isnull().sum())

print("\nDuplicate Rows")
print("Orders:", orders.duplicated().sum())
print("Items:", items.duplicated().sum())
print("Payments:", payments.duplicated().sum())

```

```

Missing Values - Orders
order_id    0
customer_id 0
order_status    0
order_purchase_timestamp    0
order_approved_at    160
order_delivered_carrier_date    1783
order_delivered_customer_date    2965
order_estimated_delivery_date    0
dtype: int64

```

```

Missing Values - Items
order_id    0
order_item_id    0
product_id    0
seller_id    0
shipping_limit_date    0
price    0
freight_value    0
dtype: int64

```

Missing Values - Payments

order_id	0
payment_sequential	0
payment_type	0
payment_installments	0
payment_value	0
dtype: int64	

Duplicate Rows

Orders: 0

Items: 0

Payments: 0